Derivatives

What is Derivative?

- Derivative is a financial instrument or security whose payoffs depend on a more primitive or fundamental good. Examples: futures, forwards, swaps, options etc.
- The underlying assets include stocks, currencies, interest rates, commodities, debt instruments, electricity prices, insurance payouts, the weather, etc.
- Financial derivative is a financial instrument whose payoffs depend on the financial instruments or security

Forward and Futures

- **Forwards**: A forward contract is a customized contract between two entities, where settlement takes place on a specific date in the future at today’s pre-agreed price.
- **Futures**: A futures contract is an agreement between two parties to buy or sell an asset at a certain time in the future at a certain price. Futures contracts are special types of forward contracts in the sense that the former are standardized exchange-traded contracts

Options

- Options are of two types – call option and put option
- Call Option: It give the holder the right but not the obligation to buy a given quantity of the underlying asset, at a given price on or before a given future date.
- Put Option: give the holder the right to sell a given quantity of the underlying asset at a given price on or before a given date.
- An American option can be exercised at any time during its life
- A European option can be exercised only at maturity

Options Cont…

- Every option has an option price, an exercise price, and an exercise date.
  - The price paid by the buyer to the writer is referred to as the option premium.
  - The exercise price or strike price is the price specified in the option contract at which the underlying asset can be purchased or sold.

Positions of the Buyer and Seller in Call and Put Options

Option Type	Buyers of Option (Long Position)	Writer of Option (Short Position)
Call	Right to Buy Asset	Obligation to Sell Asset
Put	Right to Sell Asset	Obligation to Buy Asset

Concept of Money ness

Condition	Call Option	Put Option
$S_0 > E$	In-the-Money	Out-of-the Money
$S_0 < E$	Out-of-the Money	In-the-Money
$S_0 = E$	At-the-Money	At-the-Money

Intrinsic Value and Time Value

- Option Premium = Intrinsic Value (Parity Value) + Time value (Premium over parity)
- Intrinsic value refers to the amount by which it is in-the-money
- Option which is out-of-the money has a zero intrinsic values
- For a call option which is in the money, the intrinsic value is the excess of stock price over the exercise price
- For a put option which is in the money, the intrinsic value is the excess of exercise price over the stock price

Example

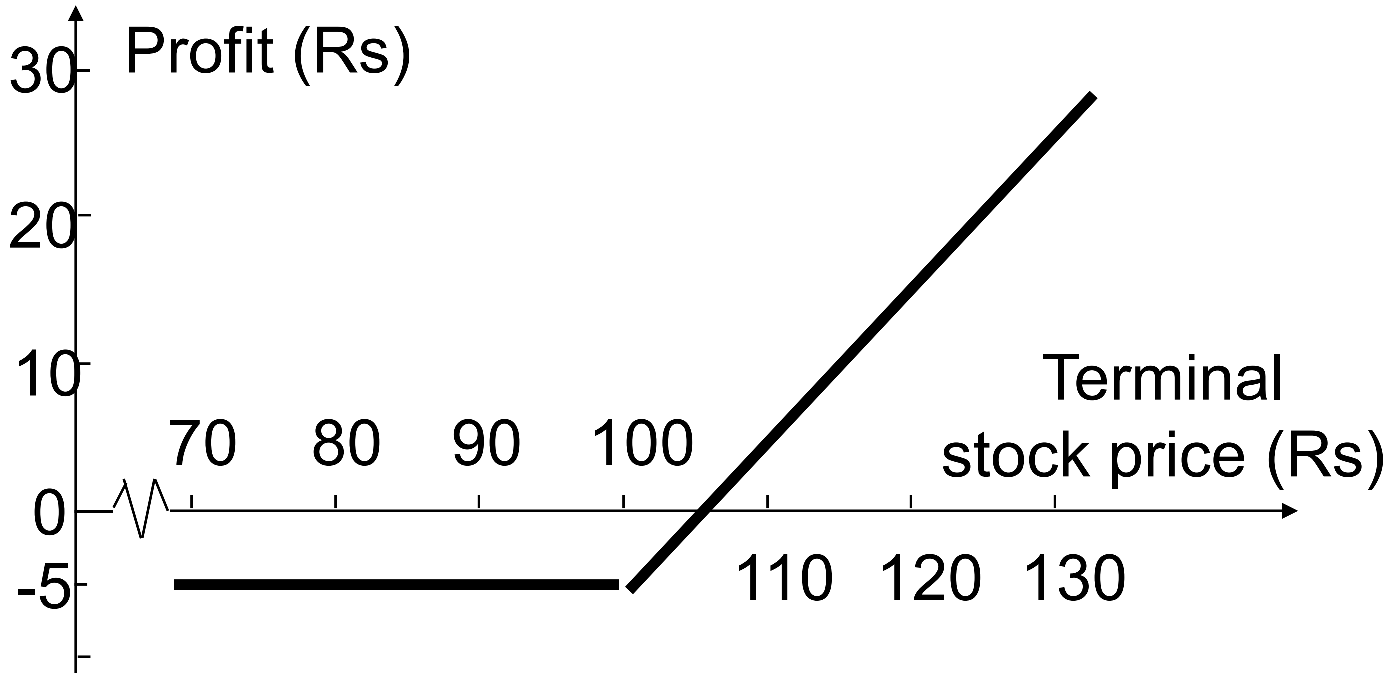
Option	Exercise price	Stock price	Call Option price	Classification	Intrinsic value	Time Value
1	80	83.5	6.75	In-the-money	3.5	6.75-3.5
2	85	83.5	2.5	Out-of the money	0	2.5

Call and Put Options at Expiration

- If the price of the underlying asset is lower than the exercise price on the expiration of a call option, the call would expire unexercised.
- When at expiration the price of the underlying asset is greater than the exercise price, the put will expire unexercised.

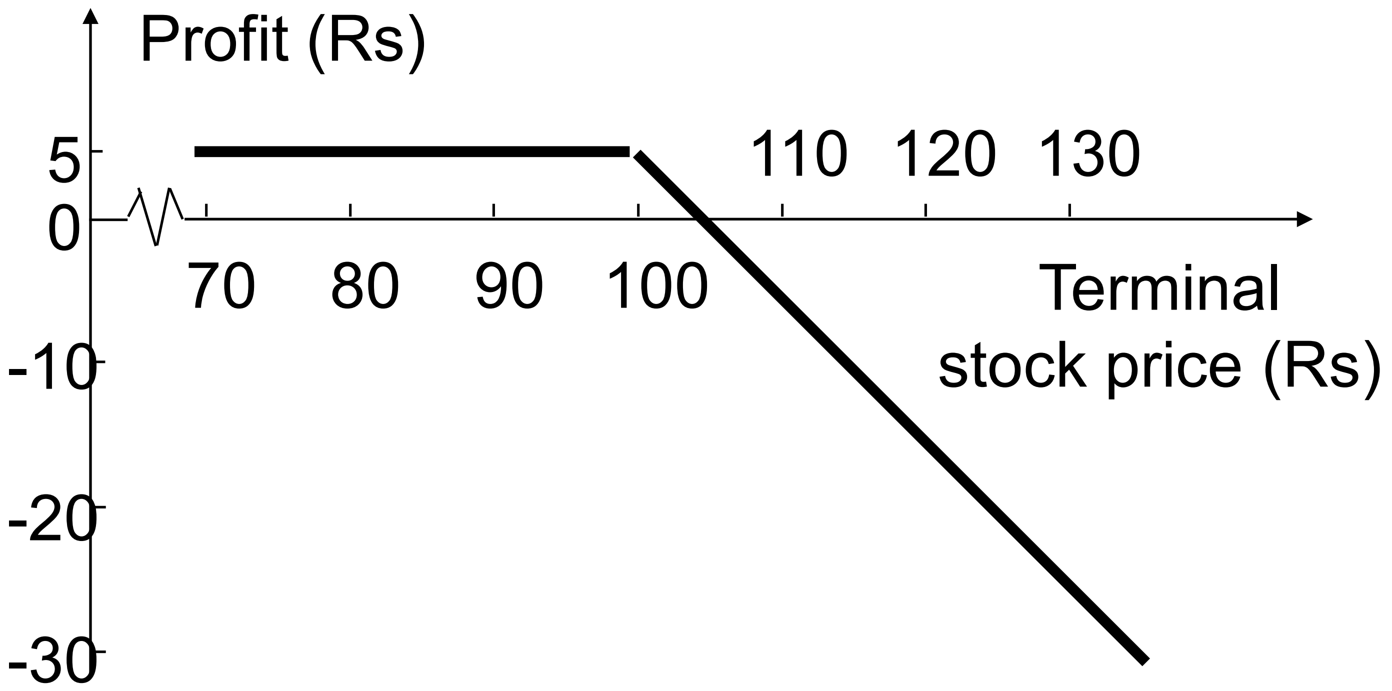
Long Call

Profit from buying one European call option: option price = Rs. 5, strike price = Rs. 100, option life = 2 months



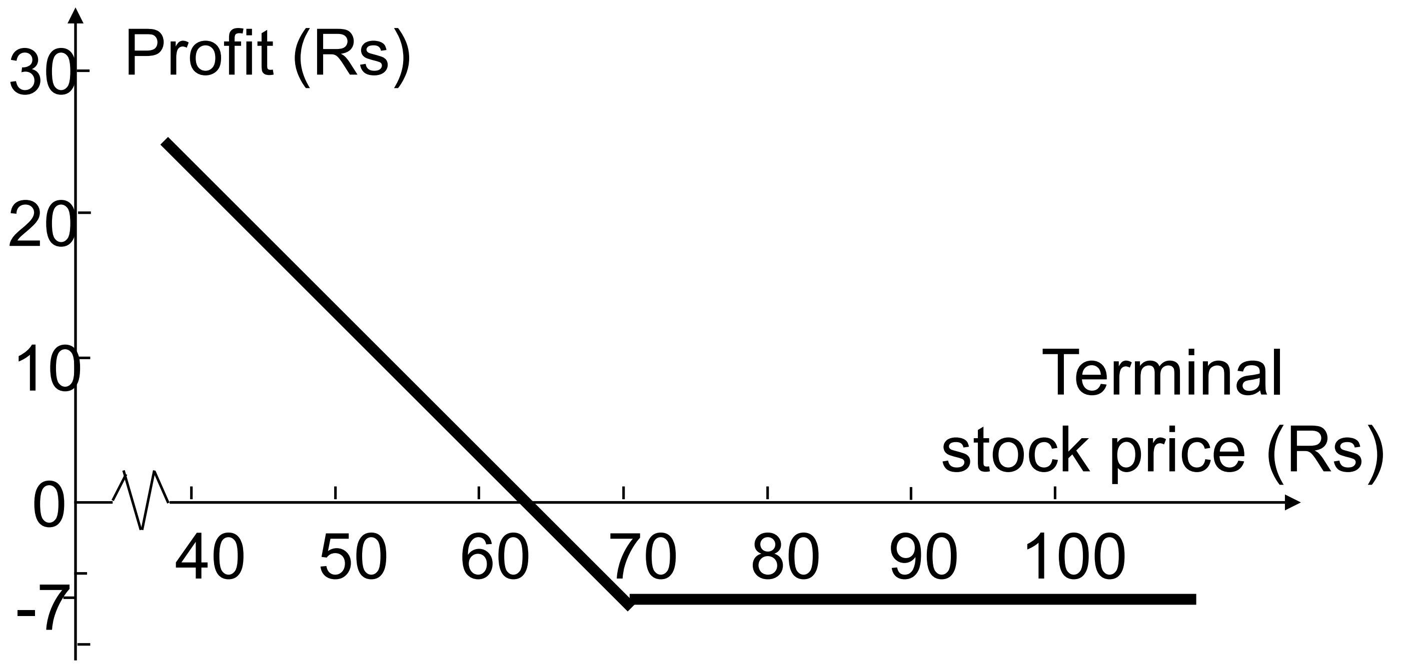
Short Call

Profit from writing one European call option: option price = Rs.5, strike price = Rs. 100



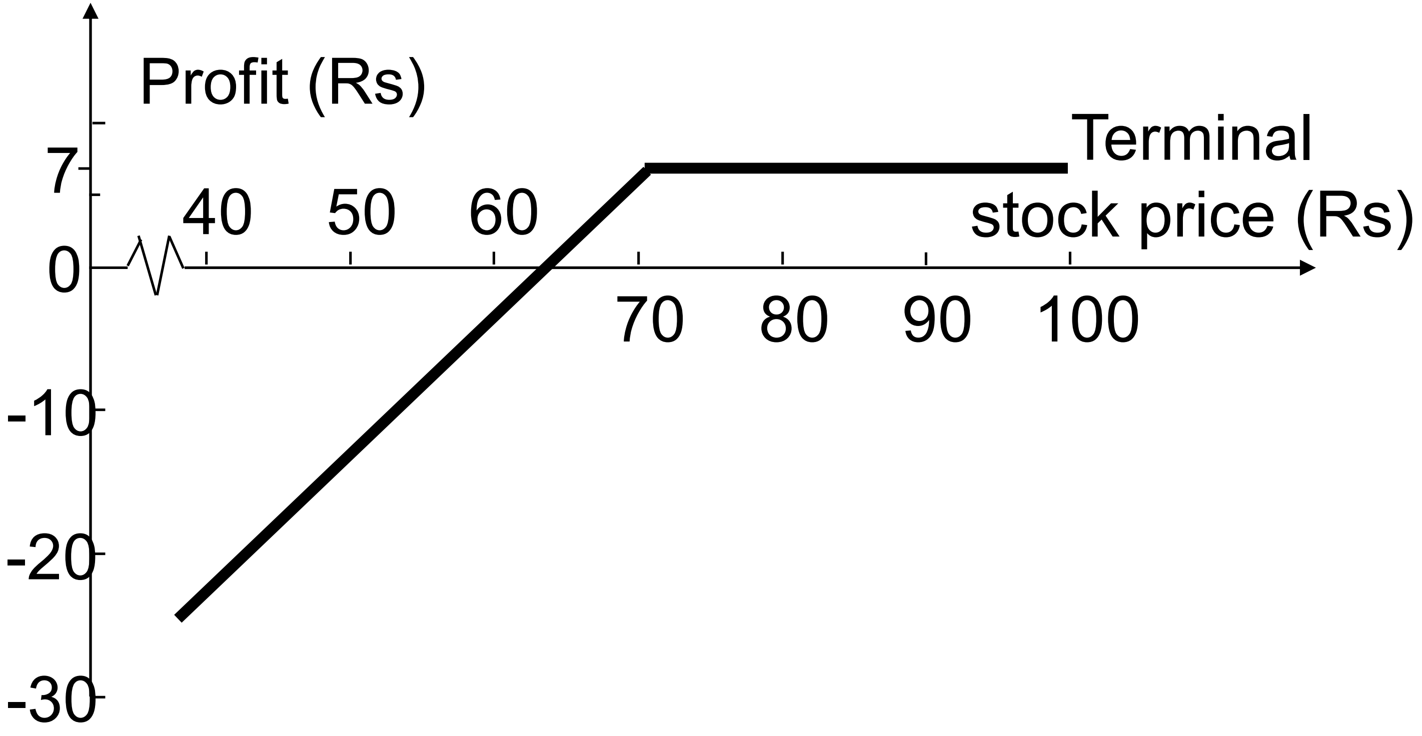
Long Put

Profit from buying a European put option: option price = Rs.7, strike price = Rs.70



Short Put

Profit from writing a European put option: option price = Rs.7, strike price = Rs.70



Swaps

- **Swaps**: Swaps are private agreements between two parties to exchange cash flows in the future according to a prearranged formula. They can be regarded as portfolios of forward contracts. The two commonly used swaps are :
- ***Interest rate swaps***: These entail swapping only the interest related cash flows between the parties in the same currency.
- ***Currency swaps***: These entail swapping both principal and interest between the parties, with the cashflows in one direction being in a different currency than those in the opposite direction.

Some Concepts

- Buyer is said to have a **long position** and seller has a **short position**
- The act of buying is called **going long** and the act of selling is called **going short**
- When one trader buys and other sells a forward contract, the transaction generates on contract of **trading volume**
- At any moment in time, there is some number of future contracts obligated for delivery and this number is called **open interest**
- **Settlement price**: the price just before the final bell each day
  - used for the daily settlement process

Forward Contracts vs Futures Contracts

FORWARDS	FUTURES
Private contract between 2 parties	Exchange traded
Not standardized	Standard contract
Usually 1 specified delivery date	Range of delivery dates
Settled at maturity	Settled daily
Delivery or final cash settlement usually occurs	Contract usually closed out prior to maturity
Some credit risk	Virtually no credit risk

Forward and Future Prices

Objective

- How forward and future prices are related to spot prices?
- Forward contracts are easy to analyse as there is no daily settlement and a single payment at maturity

Notation for Valuing Futures and Forward Contracts

S_0: Spot price today

F_0: Futures or forward price today

T: Time until delivery date

r: Risk-free interest rate for maturity T

Forward Price: Securities Providing No Income

If the spot price of is S & the futures price is for a contract deliverable in T years is F, then

$$F = S\,(1+r\,)^T$$

where r is the 1-year risk-free rate of interest.

In our examples, S=390, T=1, and r=0.05 so that

$$F = 390(1+0.05) = 409.50$$

When Interest Rates are Measured with Continuous Compounding

$$F_0 = S_0 e^{rT}$$

This equation relates the forward price and the spot price for any investment asset that provides no income and has no storage cost.

If $F_0 > S_0e^{rT}$: Investor may buy the asset by borrowing an amount equal to S_0 for a period of T at the risk free rate, and take a short position in forward contract. At the time of maturity, the assets will be delivered for a price of F and amount borrowed will be repaid by paying an amount equal to S_0e^{rT} and the deal would result in a net profit of $F_0 - S_0e^{rT}$

If $F_0 < S_0e^{rT}$: Investor would short the assets, invest the proceeds for the time period T at an interest arte r and long a forward contract. When the contract matures the asset would be purchased for a price of F and the short position in the asset would be closed out. This would result in a profit of $S_0e^{rT} - F_0$

An Arbitrage Opportunity?

- Suppose that:
  - The spot price of a non-dividend-paying stock is Rs. 40
  - The 3-month forward price is Rs. 43
  - The 3-month interest rate is 5% per annum
- Is there an arbitrage opportunity?

Suppose that:

- The spot price of nondividend-paying stock is Rs. 40
- The 3-month forward price is Rs. 39
- The 1-year interest rate is 5% per annum (continuously compounded)

Is there an arbitrage opportunity?

Example

Forward Price: 43	Forward Price: 39
Action Now:	**Action Now:**
Borrow Rs. 40 at 5% for 3 months	Short 1 unit of asset to realize Rs. 40
Buy one unit of asset	Invest Rs. 40 at 5% for 3 months
Enter into the forward contract to sell asset in 3 months for Rs. 43	Enter into a forward contract to buy asset in 3 months for Rs. 39
Action in 3 months	**Action in 3 months**
Sell asset for Rs. 43	Buy asset for Rs. 39. Close the short position
Use Rs. 40.50 to repay loan and interest	Receive Rs. 40.50 from investment
Profit: Rs. 2.50	Profit: Rs. 1.50

Example (Income is known)

- Consider a forward contract to purchase a coupon bearing bond
- Current price of coupon bearing bond: Rs. 900
- Maturity Period: 9 months
- Coupon payment in 4 months: Rs. 40
- 4-month and 9-month risk free rate: 3% and 4% per annum
- Suppose the forward price: Rs. 910

What the arbitragers will do?

Forward Price: Rs. 910	Forward Price: Rs. 870
Action Now	**Action Now**
Borrow Rs. 900 (Rs. 39.60 for 4 months and Rs. 860.40 for 9 months) to buy the bond The present value of coupon payment: $40e^{-0.03*4/12}$ = Rs. 39.60	Short 1 unit of asset to realize Rs. 900 Invest Rs. 39.60 for 4 months and Rs. 860.40 for 9 months
Buy 1 unit of asset Enter into the forward contract to sell the bond for Rs. 910	Enter into the forward contract to buy asset in 9 months for Rs. 870
Action after 4 months	**Action after 4 months**
Receive Rs. 40 as coupon	Receive Rs. 40 from 4 months investment
Use Rs. 40 to repay first loan with interest	Pay Rs. 40 on asset
Action after 9 months	**Action after 9 months**
Sell asset for Rs. 910	Receive Rs. 886.60 from 9 months investments
Use $860.40e^{0.04*0.75}$ =Rs. 886.60 to repay second loan with interest	Buy asset for Rs. 870
Profit: Rs. 23.40	Profit: Rs. 16.60

When an Investment Asset Provides a Known Income (preference Share, coupon bearing bonds)

$$F_0 = (S_0 - I)e^{rT}$$

where I is the present value of the income from investment during the life of future contract

Example: S_o =Rs. 900, I = $40e^{-0.03*4/12}$ = Rs. 39.60, r=0.04 and T= 0.75

So: $F_0 = (900-39.60)e^{0.04*0.75}$ = Rs. 886.60

If $F_0 > (S_0 - I)e^{rT}$ an arbitrager can lock in a profit by buying the asset and shorting the a forward contract on the asset.

If $F_0 < (S_0 - I)e^{rT}$ an arbitrager can lock in a profit by shorting the asset and taking a long position in a forward contract

When an Investment Asset Provides a Known Yield

when asset provides a known yield

The income is known when expressed as a percentage of the asset's price at the time income is paid.

when $F > S_0 e^{(r-q)T}$ → buys stocks underlying index & sells futures

when $F < Se^{(r-q)T}$ → buys futures & shorts or sells stocks underlying the index

$$F_0 = S_0 e^{(r-q)T}$$

where q is the average yield during the life of the contract (expressed with continuous compounding)

Example: Let: asset price: Rs. 25, Risk free rate: 10% and T=0.5, Income: 2% of the asset price once during 6 months period.

In continuous compounding it will be: 2(ln(1+0.04/2)= 3.96% i.e. q=0.0396

$F_0 = 25e^{(0.10-0.0396)*0.5}$ = Rs. 25.77

Valuing a Forward Contract

- A forward contract is worth zero when it is first negotiated
- Later it may have a positive or negative value
- Suppose that *K* is the delivery price and F_0 is the forward price for a contract that would be negotiated today
- By considering the difference between a contract with delivery price *K* and a contract with delivery price F_0 we can deduce that:
  - the value of a long forward contract is

$$(F_0 - K)e^{-rT}$$

  - the value of a short forward contract is

$$(K - F_0)e^{-rT}$$

Futures and Forwards on Currencies

- A foreign currency is analogous to a security providing a yield
- The yield is the foreign risk-free interest rate
- It follows that if r_f is the foreign risk-free interest rate

$$F_0 = S_0 e^{(r - r_f)T}$$

- This is the well-known interest rate parity relationship

Option Valuation
Upper and Lower Bounds for Option Prices

- If an option price is above the upper bound and below the lower bound, there are profitable opportunities for arbitrageurs.
- Upper Bounds (Call Option): $c \leq S_0$ and $C \leq S_0$, if these relationship does not hold then an arbitrageurs can easily make risk less profit by buying the stock and selling the call option
- Upper Bound (Put Option): $p \leq K$ and $P \leq K$, for European option $p \leq Ke^{-rt}$, If this is not true then risk less profit can be made by writing the option and investing the proceeds of the sale at the risk-free interest rate.

Lower Bounds for calls on Non-Dividend paying Stocks

$$c \geq S_0 - Ke^{-rT}$$

- Suppose that

c = 3	S_0 = 20
T = 1	r = 10%
K = 18	D = 0

- Is there an arbitrage opportunity?

Answer

- S_0- Ke^{-rt} = 3.71
- Arbitrageur can buy the call and short the stock
- Cash Inflow = 20-3 = 17
- Invest for one year at 10% per annum
- Total money = $17e^{0.1}$=Rs. 18.79
- If stock price is greater than 18 he can exercise the option to buy the stock and close the short position and profit will be : rs. 18.79 – Rs. 18 = 0.79

Lower Bound for European Puts on Non-Dividend paying Stocks

$$p \geq Ke^{-rT} - S_0$$

- Suppose that

p	= 1	S_0	= 37
T	= 0.5	r	=5%
K	= 40	D	= 0

- Is there an arbitrage opportunity?

Answer

- $Ke^{-rt} – S$ = Rs. 2.01
- It is more than the put price.
- The arbitrageurs can borrow Rs. 38 for six months to buy both put and the stock
- He is required to pay $38e^{0.05*0.5}$= rs. 38.96
- If the stock price is below 40, the arbitrageur exercises the option to sell the stock for Rs. 40 repays the loan and makes a profit of Rs 40 – Rs. 38.96 = Rs. 1.04

Put-Call Parity

- Consider the following 2 portfolios:
  – Portfolio A: European call on a stock + PV of the strike price in cash
  – Portfolio C: European put on the stock + the stock
- Both are worth MAX(S_T, K) at the maturity of the options
- They must therefore be worth the same today. This means that

$$c + Ke^{-rT} = p + S_0$$

→ If this not satisfied there is arbitrage opportunities

Arbitrage Opportunities

- Suppose that

  c = 3 S_0 = 31

  T = 0.25 r = 10%

  K =30 D = 0

- What are the arbitrage possibilities when p = 2.25 ?

Answer

- $c + Ke^{-rT} = p + S_0$
- $3 + 30e^{-0.1*3/12}$= Rs.32.26
- $P + S_0$ = 2.25 + 31 = Rs. 33.25
- Arbitrage Strategy: Buying call and shorting both put and stock generating a positive cash flow of: -3 + 2.25 +31 = Rs. 30.25
- Invest it at risk free interest rate, amount grows to $30.25e^{0.1*0.25}$=Rs.31.02
- Is stock price at expiration of option is greater than 30 the call will be exercised and if it is less than 30, then the put will be exercised.
- Net profit = 31.02 -30 = rs 1.02

The Impact of Dividends on Lower Bounds to Option Prices

$$c \geq S_0 - D - Ke^{-rT}$$

$$p \geq D + Ke^{-rT} - S_0$$

Swap

A swap is an agreement to exchange cash flows at specified future times according to certain specified rules.

Interest Rate Swap

- The most popular (plain vanilla) interest rate swap is one where LIBOR is exchanged for a fixed rate of interest.
- In an interest rate swap, one company agrees to pay to another company cash flows equal to interest at a predetermined fixed rate on a notional principal for a predetermined number of years. In return, it receives interest at a floating rate on the same notional principal for the same period of time from the other company.

Interest Rate Swap

- The floating rate in most interest rate swap agreements is the London Interbank Offered Rate (LIBOR). It is the rate of interest at which a bank with a AA credit rating is able to borrow from other banks.
- Example: consider a 10-year bond with a rate of interest specified as 6-month LIBOR plus 0.5% per annum. The life of the bond is divided into 20 periods, each 6 months in length. For each period, the rate of interest is set at 0.5% per annum above the 6-month LIBOR rate at the beginning of the period. Interest is paid at the end of the period.

Example

- Consider a hypothetical 3-year swap initiated on March 15, 2016, between Company X and Company Y.
- Suppose X agrees to pay Y an interest rate of 5% per annum on a principal of Rs.100 million, and in return Y agrees to pay X the 6-month LIBOR rate on the same principal.
- X is the fixed-rate payer; Y is the floating rate payer.
- We assume the agreement specifies that payments are to be exchanged every 6 months and that the 5% interest rate is quoted with semi-annual compounding.

Example: Cash Flows to Company X

		---------Millions of Rupees---------		
	LIBOR	*FLOATING*	*FIXED*	Net
Date	Rate	Cash Flow	Cash Flow	Cash Flow
Mar.15, 2016	4.2%			
Sept. 15, 2016	4.8%	+2.10	–2.50	–0.40
Mar.15, 2017	5.3%	+2.40	–2.50	–0.10
Sept. 15, 2017	5.5%	+2.65	–2.50	+0.15
Mar.15, 2018	5.6%	+2.75	–2.50	+0.25
Sept. 15, 2018	5.9%	+2.80	–2.50	+0.30
Mar.15, 2019	6.4%	+2.95	–2.50	+0.45

Example

- The principal itself is not exchanged. For this reason it is termed the notional principal, or just the notional.
- If the notional principal were exchanged at the end of the life of the swap, the nature of the deal would not be changed in any way.
- The notional principal is the same for both the fixed and floating payments. Exchanging Rs. 100 million for Rs. 100 million at the end of the life of the swap is a transaction that would have no financial value to either company X or Y.

Uses of an Interest Rate Swap

- Converting a liability from
  - fixed rate to floating rate
  - floating rate to fixed rate

- Converting an investment from
  - fixed rate to floating rate
  - floating rate to fixed rate

Using a swap to transform a liability

- Company X has borrowed Rs. 100 million at LIBOR+0.1%, wants to transform a floating rate loan into a fixed rate loan. After it has entered into a swap:
- Loan payment: LIBOR+0.1%
- Add:Paid under swap + 5%
- Less:Received under swap - LIBOR
- Net Payment 5.1%

Using a swap to transform a liability Con…

- For Y swap is used to transform fixed rate into floating rate. Suppose it has 3-year Rs.100 million loan on which it pays 5.2%
- Loan payment 5.2%
- Add: Paid under swap +LIBOR
- Less: Received under swap - 5%
- Net payment LIBOR+0.2%

Using the swap to transform an asset (nature of an asset)

- Suppose Company X owns Rs.100 million in bonds that will provide 4.7% per annum over the next 3 years. Now company X enters into a swap, wants to switch its assets from fixed to floating rate.
- Investment income 4.7%
- Less: Paid under swap -5%
- Add: Received under swap +LIBOR
- Net income LIBOR-0.3%

Using the swap to transform an asset (nature of an asset) Con…

- Company Y is transforming an asset earning floating to fixed. Suppose Y has an investment Rs.100 million that yields LIBOR-0.20. After it has entered into the swap:
- Investment income LIBOR-0.20
- Less: Paid under swap - LIBOR
- Add: Received under swap + 5%
- Net Investment income 4.8%

The Comparative Advantage Argument

AAA Company and BBB Company wish to borrow Rs.10 million for 5 years

- AAA wants to borrow floating
- BBB wants to borrow fixed

	Fixed	*Floating*
AAA	4.0%	6-month LIBOR − 0.10%
BBB	5.2%	6-month LIBOR + 0.6%

The Comparative Advantage Argument

- BBB has a comparative advantage in floating market and AAA has a comparative advantage in fixed rate market.
- They can enter into a swap that AAA ends up with floating rate funds and BBB ends up with fixed rate funds.
- Suppose AAA agrees to pay interest at 6 month LIBOR on Rs.10 million and BBB agrees to pay 4.35% per annum on Rs.10 million.

Comparative Advantage Argument

	AAA	BBB
Loan payment	4%	LIBOR+0.6%
Add: Paid under swap	+ LIBOR	+ 4.35%
Less: Received under swap	-4.35%	-LIBOR
Net Payment (Before)	LIBOR-0.35% (LIBOR-0.10%) (0.25% gain)	4.95% (5.2%) (0.25% gain)

Comparative Advantage Argument

- The swap agreement appears to improve the position of both company.
- Total gain is: 0.25+0.25=0.50%
- If “a” is the difference between the interest rates in fixed market and if “b’’ is the difference between the interest rates in floating market.
- Total gain is; a-b
- a = 1.2% (5.2%-5%)
- b= 0.7% (LIBOR+0.6%-(LIBOR-0.1%))
- a-b = 1.2-0.7=0.5%

Criticism of the Comparative Advantage Argument

- The 4.0% and 5.2% rates available to AAACorp and BBBCorp in fixed rate markets are 5-year rates
- The LIBOR−0.1% and LIBOR+0.6% rates available in the floating rate market are six-month rates
- BBBCorp’s fixed rate depends on the spread above LIBOR it borrows at in the future

Currency Swaps

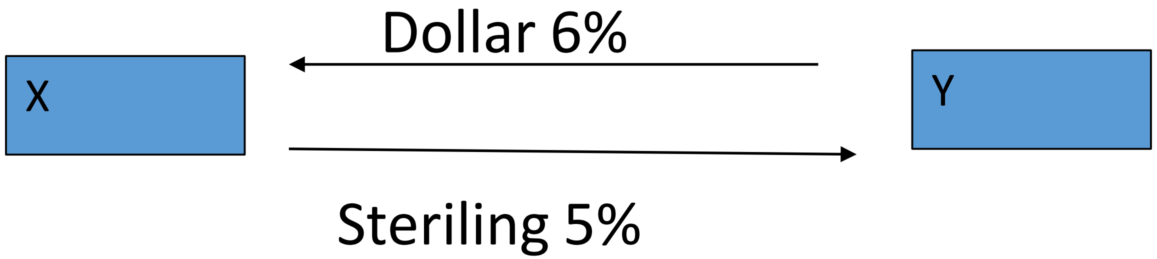
- Exchanging principal and interest payments in one currency for principal and interest payments in another currency.
- In an interest rate swap the principal is not exchanged
- In a currency swap the principal is usually exchanged at the beginning and the end of the swap’s life

Fixed-for-fixed currency swap

- This involves exchanging principal and interest payments at a fixed rate in one currency for principal and interest payments at a fixed rate in another currency.
- Usually the principal amounts are chosen to be approximately equivalent using the exchange rate at the swap’s initiation. When they are exchanged at the end of the life of the swap, their values may be quite different.

Example

- Consider a 5-year currency swap agreement between X and Y entered into on February 1, 2015. We suppose that X pays a fixed rate of interest of 5%in sterling and receives a fixed rate of interest of 6%in dollars from Y. Interest rate payments are made once a year and the principal amounts are \$15 million and £10 million. This is termed a fixed-for-fixed currency swap because the interest rate in each currency is at a fixed rate.



Cash Flow to X

	Dollar cash flow (millions)	Sterling cash flow (millions)
February 1, 2015	-15.00	+10.00
February 1, 2016	+0.90	-0.50
February 1, 2017	+0.90	-0.50
February 1, 2018	+0.90	-0.50
February 1, 2019	+0.90	-0.50
February 1, 2020	+15.9	-10.50

Typical Uses of a Currency Swap

- Conversion from a liability in one currency to a liability in another currency

- Conversion from an investment in one currency to an investment in another currency

Use of a Currency Swap to Transform Liabilities and Assets

- A swap can be used to transform borrowings in one currency to borrowings in another. Suppose that X can issue $15 million of USdollar-denominated bonds at 6% interest. The swap has the effect of transforming this transaction into one where X has borrowed £10 million at 5% interest.
- The initial exchange of principal converts the proceeds of the bond issue from US dollars to sterling. The subsequent exchanges in the swap have the effect of swapping the interest and principal payments from dollars to sterling.
- The swap can also be used to transform the nature of assets. Suppose that X can invest £10 million in the UK to yield 5% per annum for the next 5 years, but feels that the US dollar will strengthen against sterling and prefers a US-dollar-denominated investment. The swap has the effect of transforming the UK investment into a $15 million investment in the US yielding 6%.

Comparative Advantage Arguments for Currency Swaps

General Electric wants to borrow 20 miilion AUD and Qantas wants to borrow 15 million USD (Exchange rate is $0.75 per AUD)

	USD	AUD
General Motors	5.0%	7.6%
Qantas	7.0%	8.0%

GE has a comparative advantage in USD and Qantas has a comparative advantage in AUD. So that GE borrows USD and Qantas borrows AUS then they enter into a currency swap to transform GE's loan into a AUD and Qantas loan into a USD.

Total gain to all parties: 2%-0.4% = 1.6%